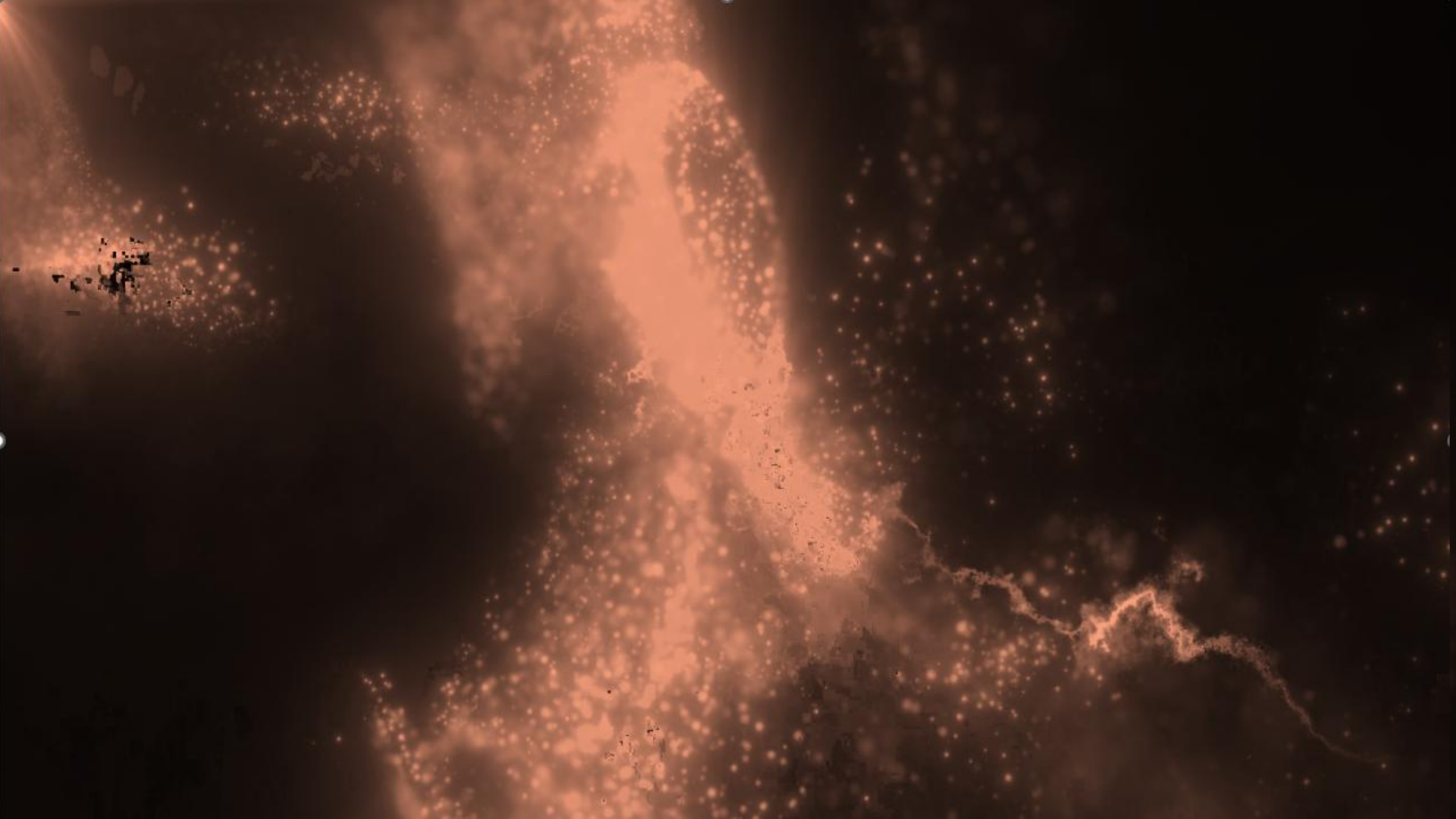
SUBSCRIBE



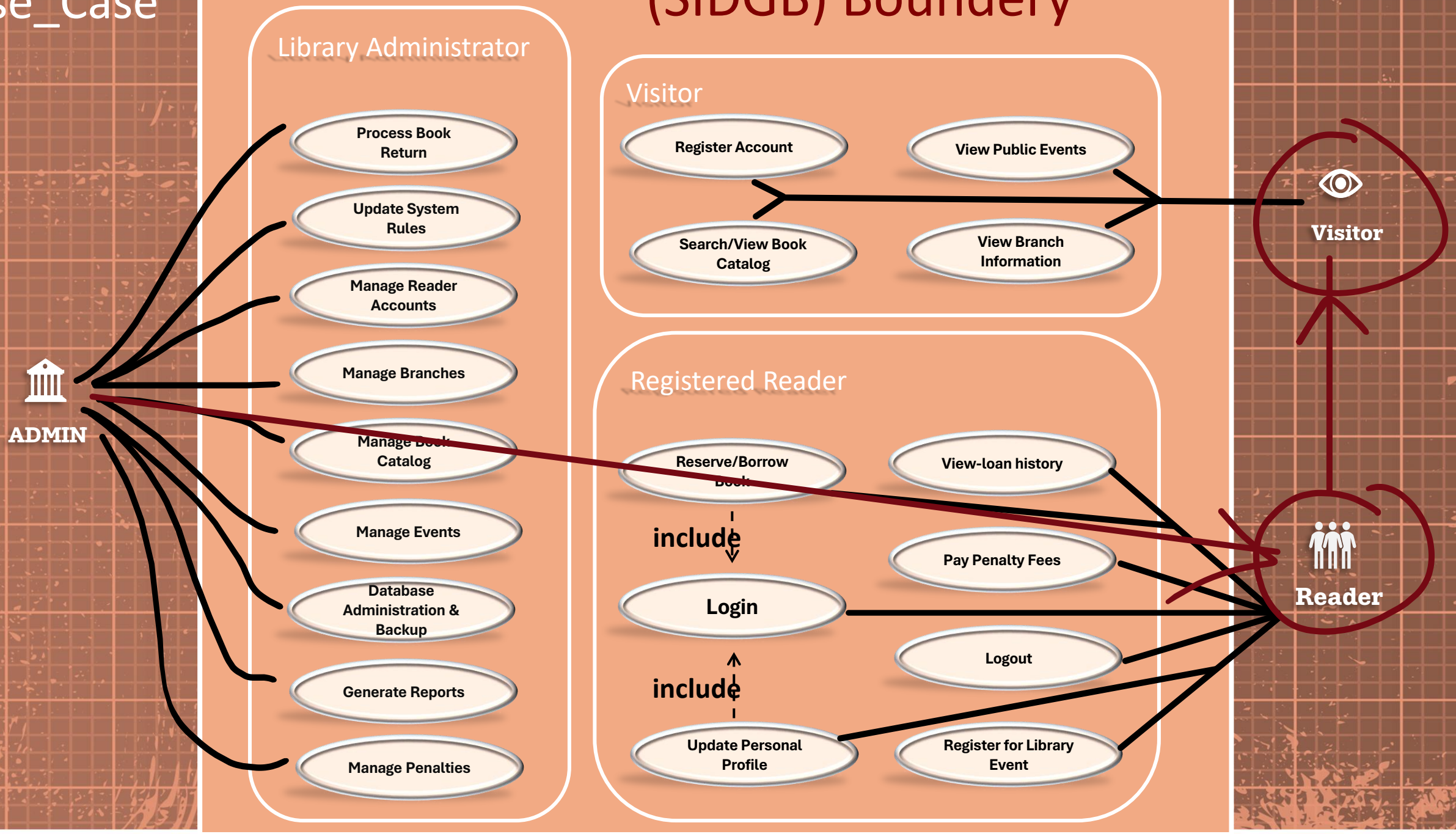
Reader

ADMIN

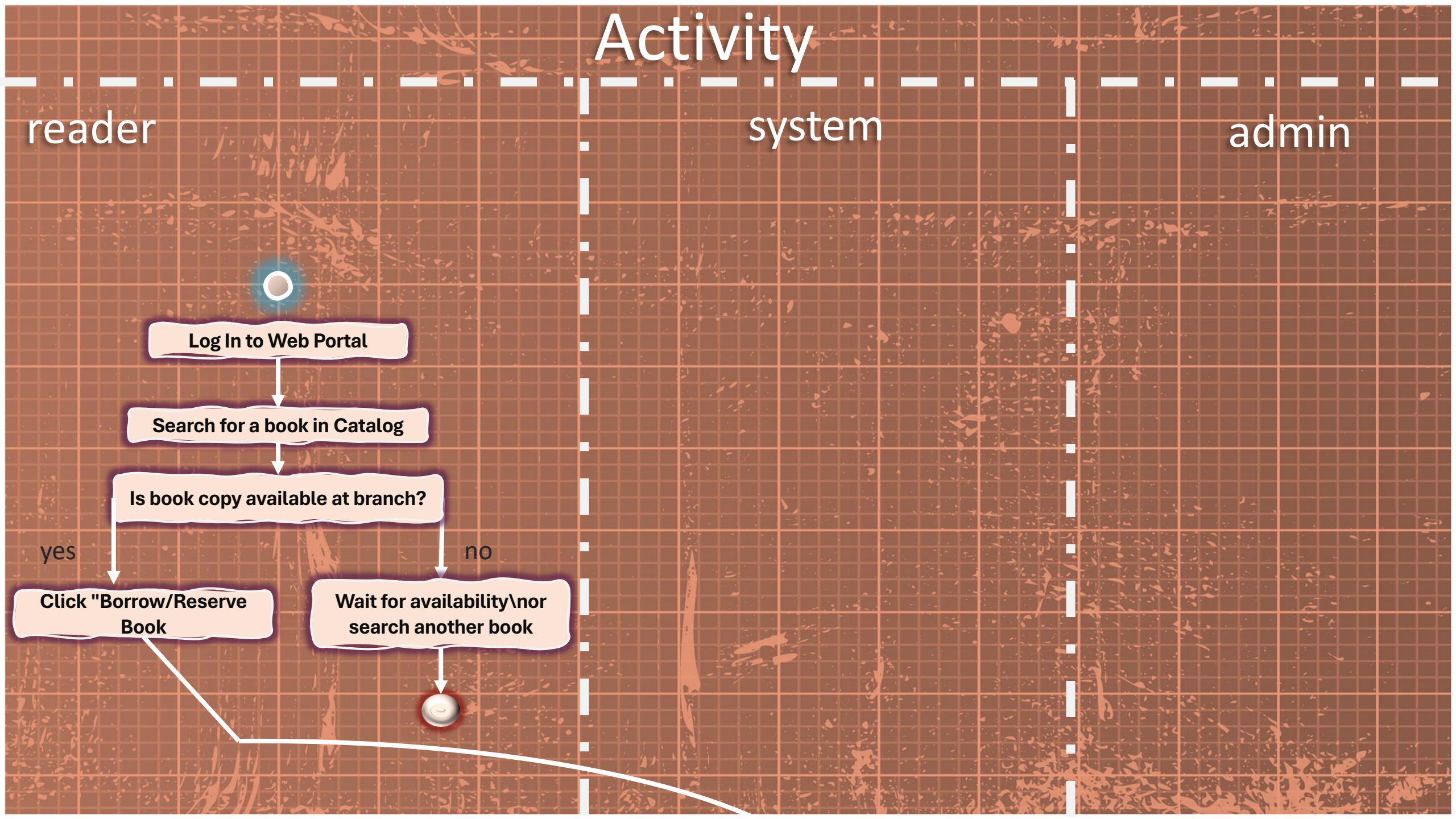
Visitor

Use_Case

(SIDGB) Boundery



Activity


reader
system
admin
Log In to Web Portal
Search for a book in Catalog
Is book copy available at branch?
yes
no
Click "Borrow/Reserve Book
Wait for availability\nor search another book

Activity

reader | system | admin

Verify Reader status
(Check if in "Unwanted Clients

Is Reader in Blacklist?

Display Error Message “Borrowing Suspended"

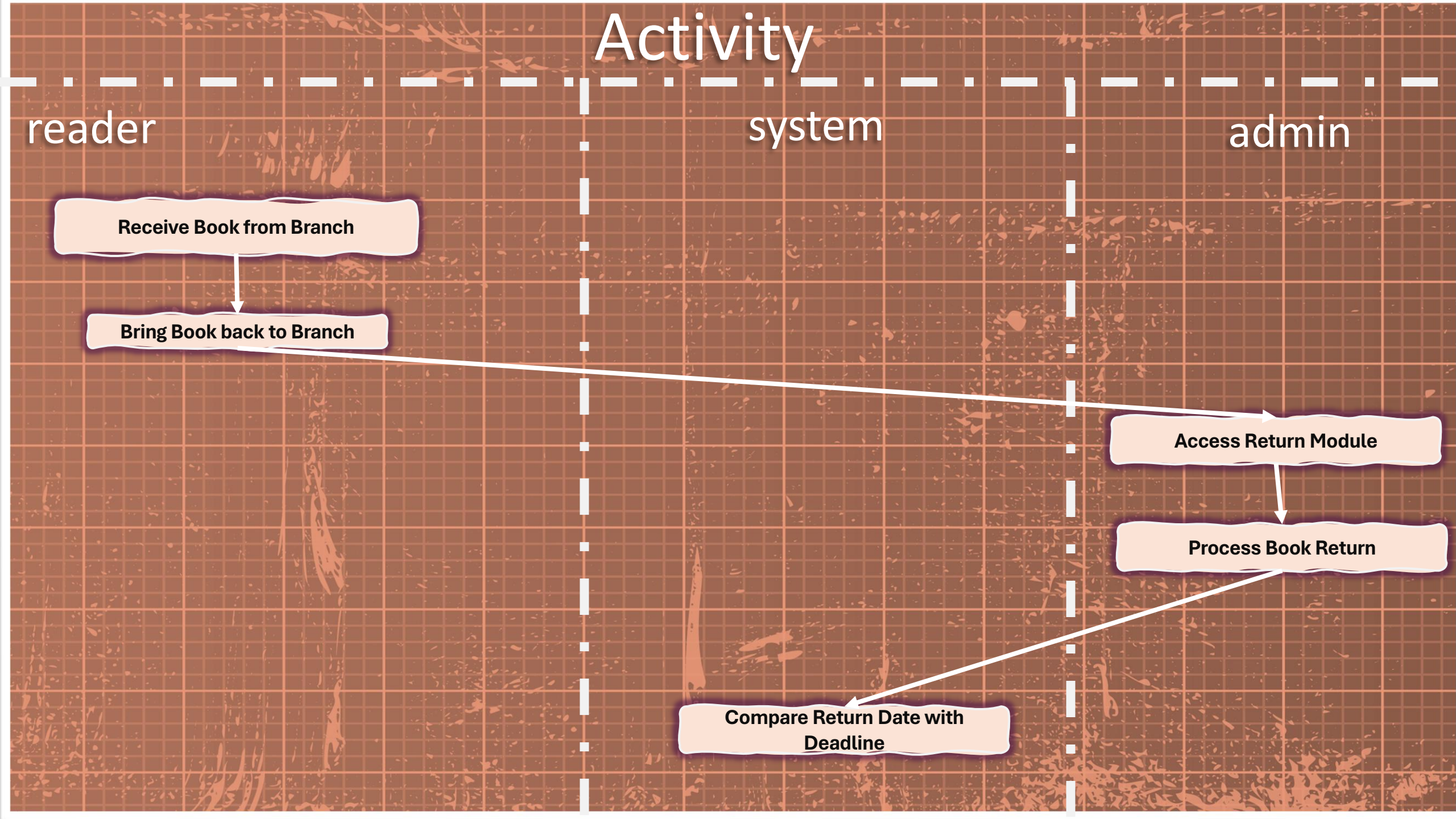
Register Loan Date

Is book copy available at branch?

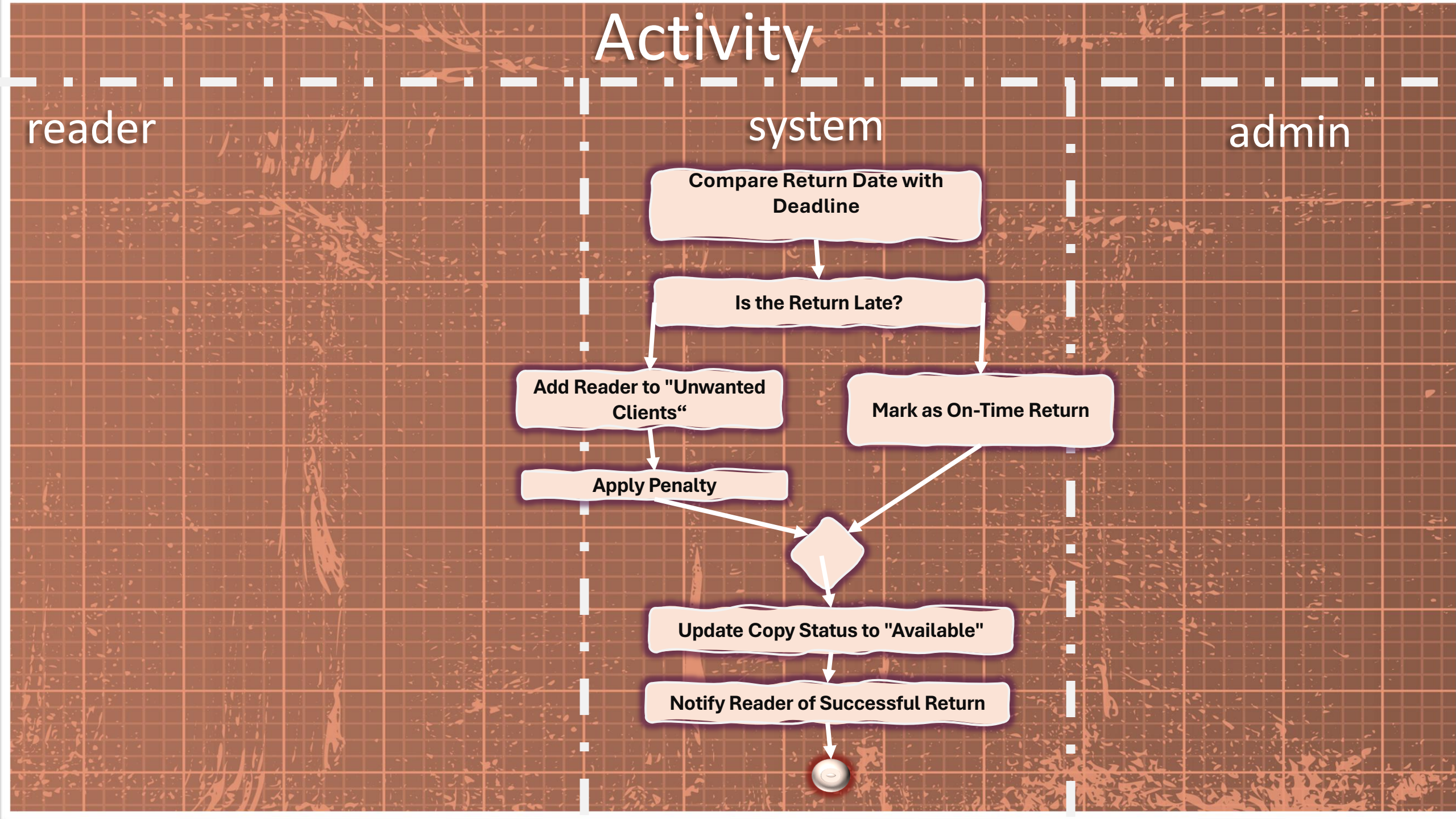
Calculate Return Deadline (14 days)

Update Copy Status to "Borrowed"

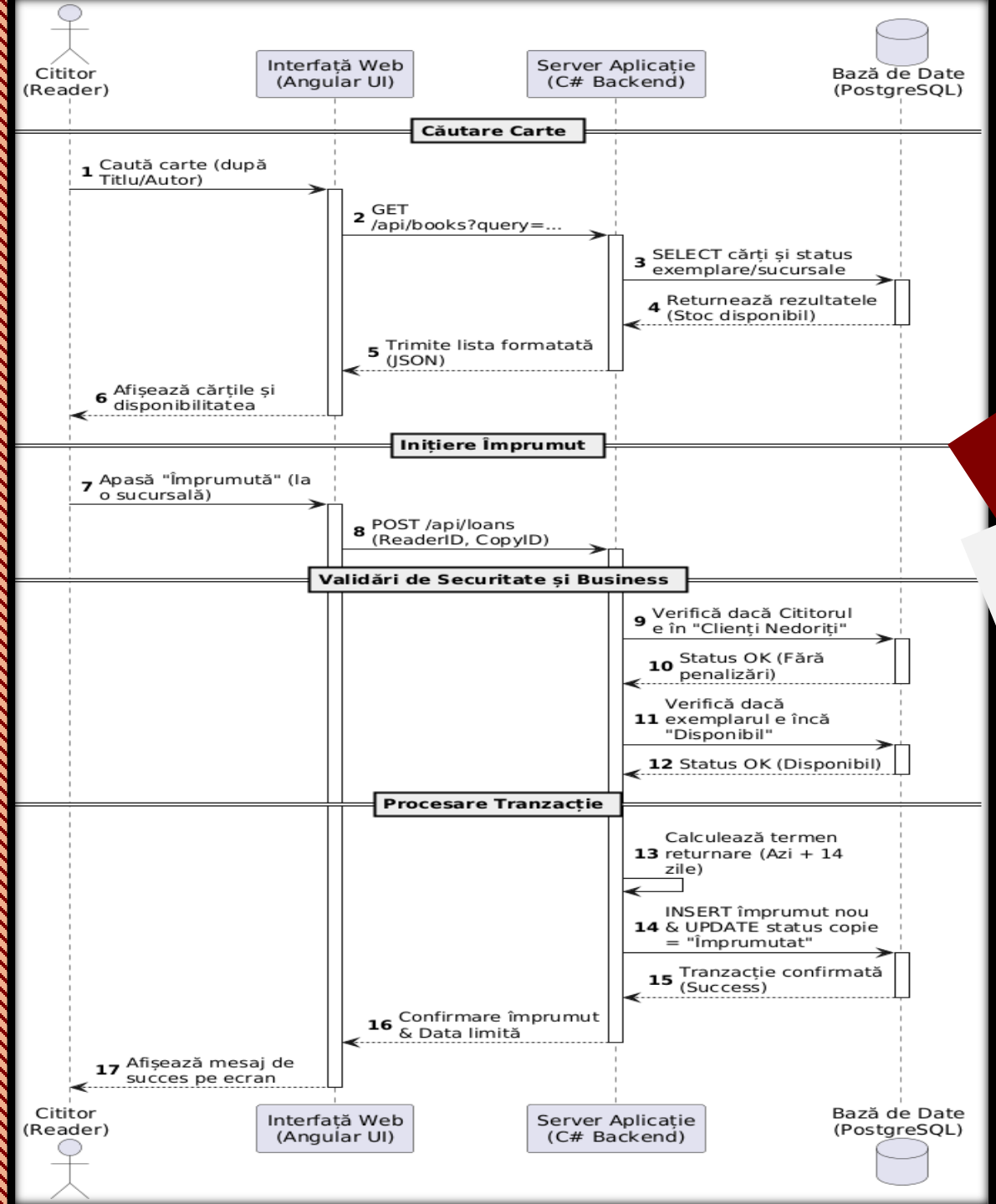
Activity

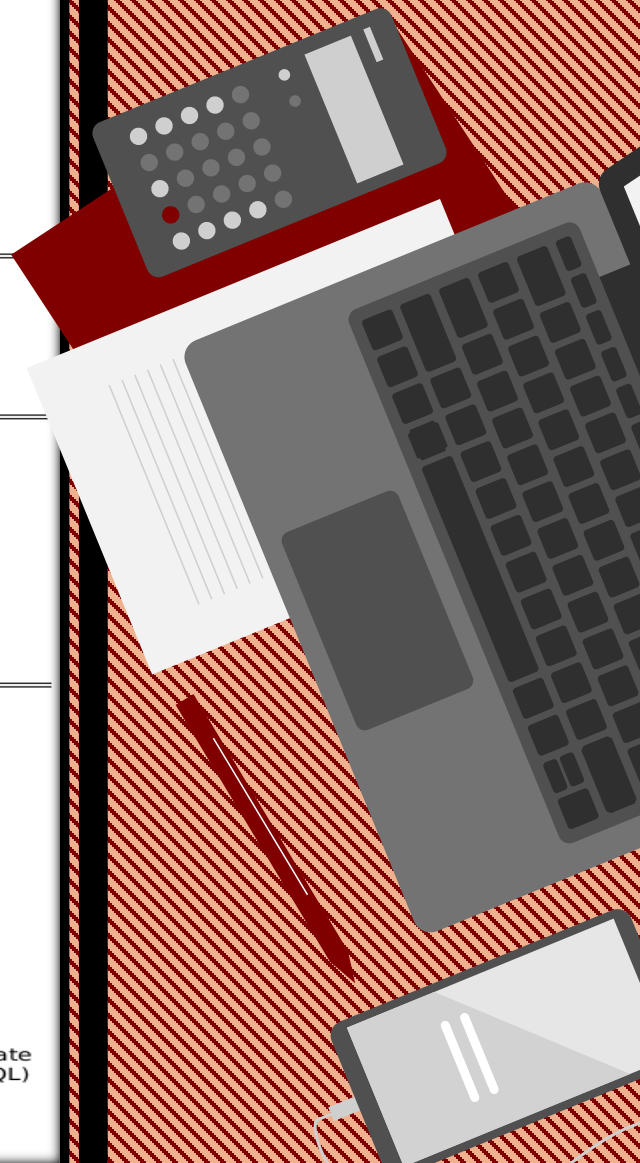

reader
system
admin
Receive Book from Branch
Bring Book back to Branch
Access Return Module
Process Book Return
Compare Return Date with Deadline

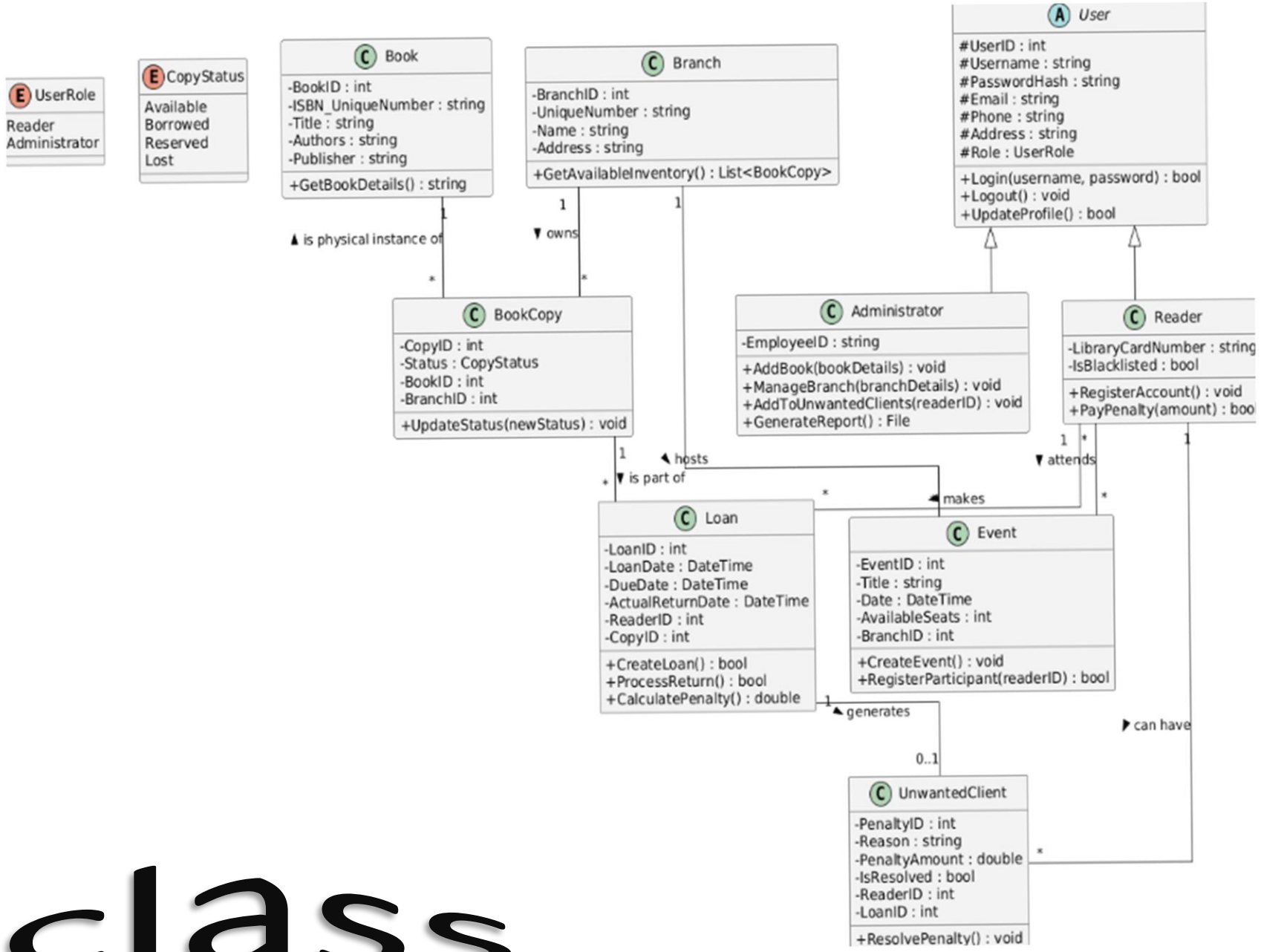
Activity


reader
system
admin
Compare Return Date with Deadline
Is the Return Late?
Add Reader to "Unwanted Clients“
Mark as On-Time Return
Apply Penalty
Update Copy Status to "Available"
Notify Reader of Successful Return

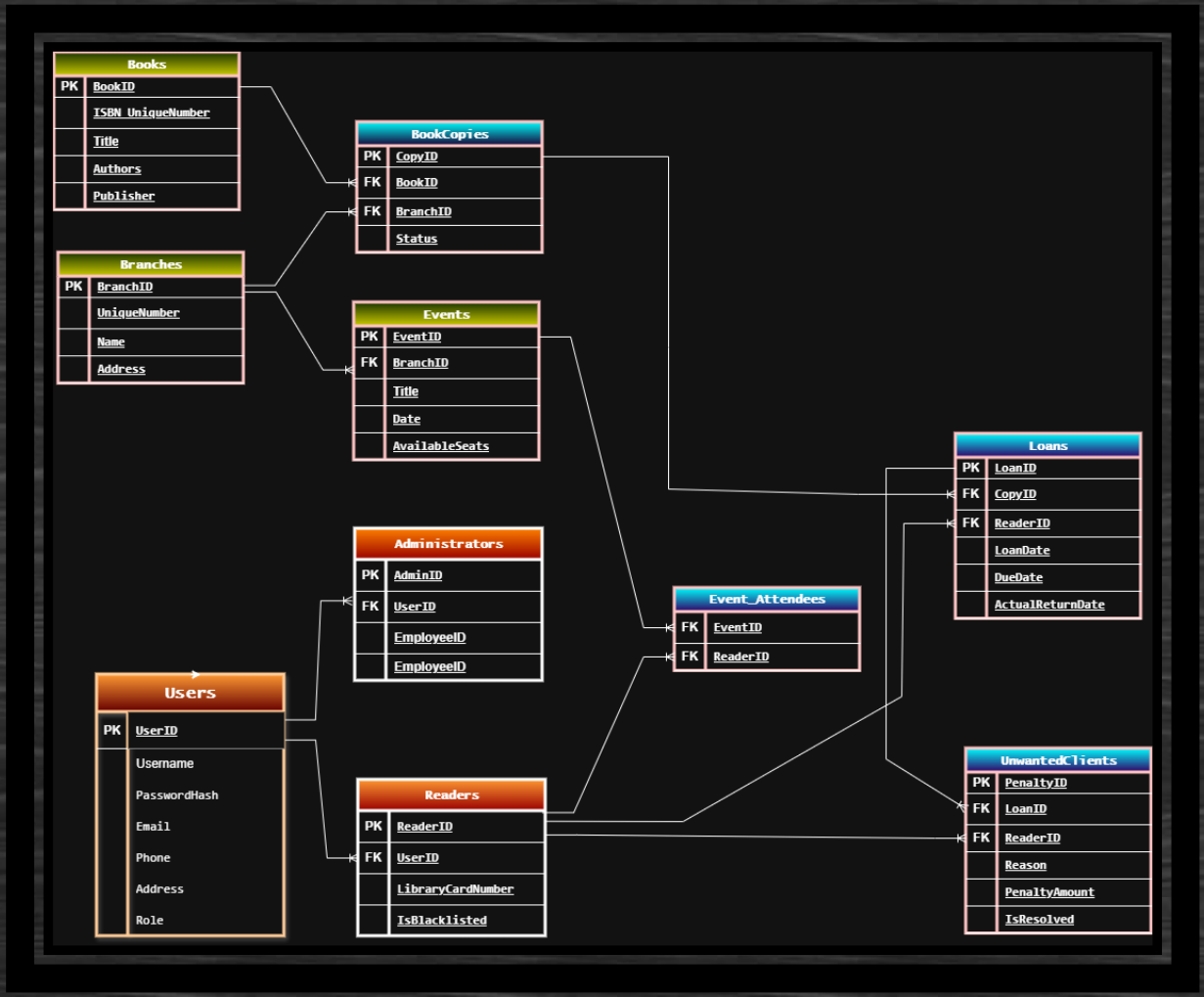
SEQUENCE


Cititor (Reader)
Interfață Web (Angular UI)
Server Aplicație (C# Backend)
Bază de Date (PostgreSQL)
Căutare Carte
1 Caută carte (după Titlu/Autor)
2 GET /api/books?query=...
3 SELECT cărți și status exemplare/sucursale
4 Returnează rezultatele (Stoc disponibil)
5 Trimite lista formatată (JSON)
6 Afișează cărțile și disponibilitatea
Inițiere Împrumut
7 Apasă "Împrumută" (la o sucursală)
8 POST /api/loans (ReaderID, CopyID)
Validări de Securitate și Business
9 Verifică dacă Cititorul e în "Clienți Nedoriți"
10 Status OK (Fără penalizări)
11 Verifică dacă exemplarul e încă "Disponibil"
12 Status OK (Disponibil)
Procesare Tranzacție
13 Calculează termen returnare (Azi + 14 zile)
14 INSERT împrumut nou & UPDATE status copie = "Împrumutat"
15 Tranzacție confirmată (Success)
16 Confirmare împrumut & Data limită
17 Afișează mesaj de succes pe ecran
Cititor (Reader)
Interfață Web (Angular UI)
Server Aplicație (C# Backend)
Bază de Date (PostgreSQL)




UserRole
Reader
Administrator
CopyStatus
Available
Borrowed
Reserved
Lost
Book
-BookID : int
-ISBN_UniqueNumber : string
-Title : string
-Authors : string
-Publisher : string
+GetBookDetails() : string
Branch
-BranchID : int
-UniqueNumber : string
-Name : string
-Address : string
+GetAvailableInventory() : List<BookCopy>
User
#UserID : int
#Username : string
#PasswordHash : string
#Email : string
#Phone : string
#Address : string
#Role : UserRole
+Login(username, password) : bool
+Logout() : void
+UpdateProfile() : bool
is physical instance of
owns
BookCopy
-CopyID : int
-Status : CopyStatus
-BookID : int
-BranchID : int
+UpdateStatus(newStatus) : void
Administrator
-EmployeeID : string
+AddBook(bookDetails) : void
+ManageBranch(branchDetails) : void
+AddToUnwantedClients(readerID) : void
+GenerateReport() : File
Reader
-LibraryCardNumber : string
-IsBlacklisted : bool
+RegisterAccount() : void
+PayPenalty(amount) : boo
hosts
is part of
attends
makes
Loan
-LoanID : int
-LoanDate : DateTime
-DueDate : DateTime
-ActualReturnDate : DateTime
-ReaderID : int
-CopyID : int
+CreateLoan() : bool
+ProcessReturn() : bool
+CalculatePenalty() : double
Event
-EventID : int
-Title : string
-Date : DateTime
-AvailableSeats : int
-BranchID : int
+CreateEvent() : void
+RegisterParticipant(readerID) : bool
generates
can have
0..1
UnwantedClient
-PenaltyID : int
-Reason : string
-PenaltyAmount : double
-IsResolved : bool
-ReaderID : int
-LoanID : int
+ResolvePenalty() : void

class


Books
PK BookID
ISBN_UniqueNumber
Title
Authors
Publisher
BookCopies
PK CopyID
FK BookID
FK BranchID
Status
Branches
PK BranchID
UniqueNumber
Name
Address
Events
PK EventID
FK BranchID
Title
Date
AvailableSeats
Loans
PK LoanID
FK CopyID
FK ReaderID
LoanDate
DueDate
ActualReturnDate
Administrators
PK AdminID
FK UserID
EmployeeID
EmployeeID
Event_Attendees
FK EventID
FK ReaderID
Users
PK UserID
Username
PasswordHash
Email
Phone
Address
Role
Readers
PK ReaderID
FK UserID
LibraryCardNumber
IsBlacklisted
UnwantedClients
PK PenaltyID
FK LoanID
FK ReaderID
Reason
PenaltyAmount
IsResolved

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